

3. ATP is described as the universal energy currency of cells.

(a) (i) Describe why ATP is described as a universal currency. [2]

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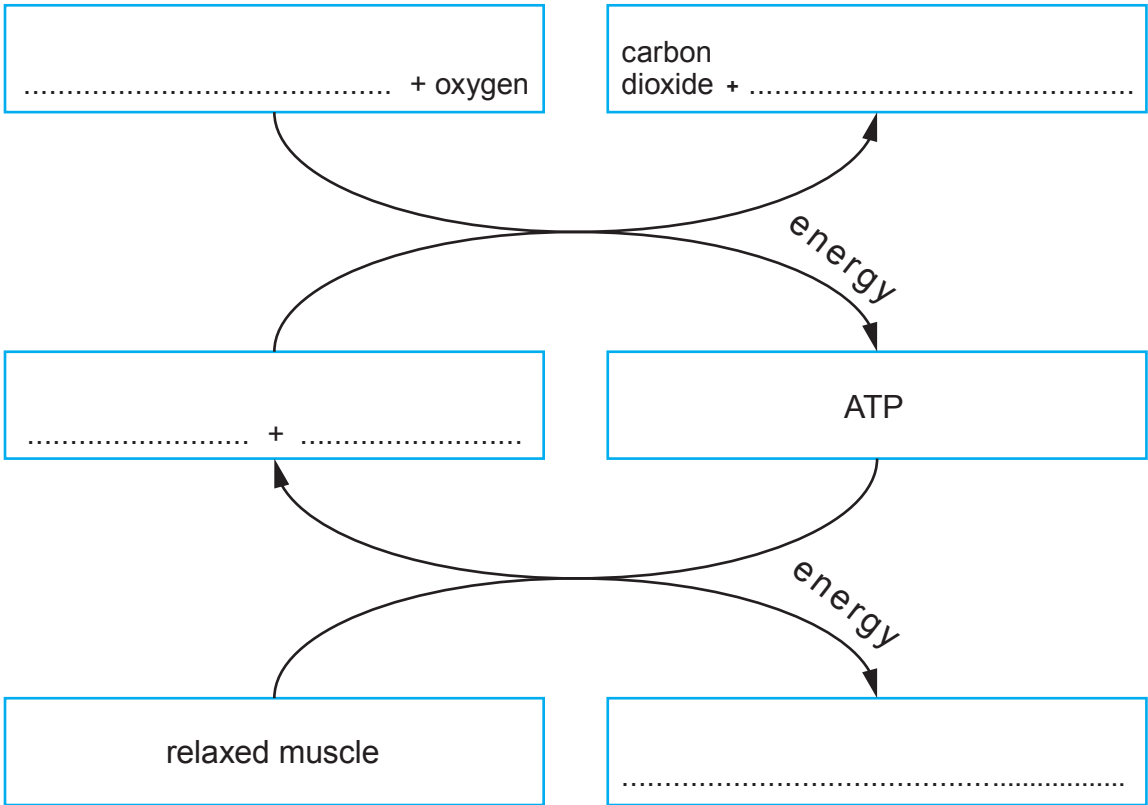
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Image 3.1 summarises the production and use of ATP in muscle cells.

(ii) Complete the diagram in Image 3.1. [3]

Image 3.1

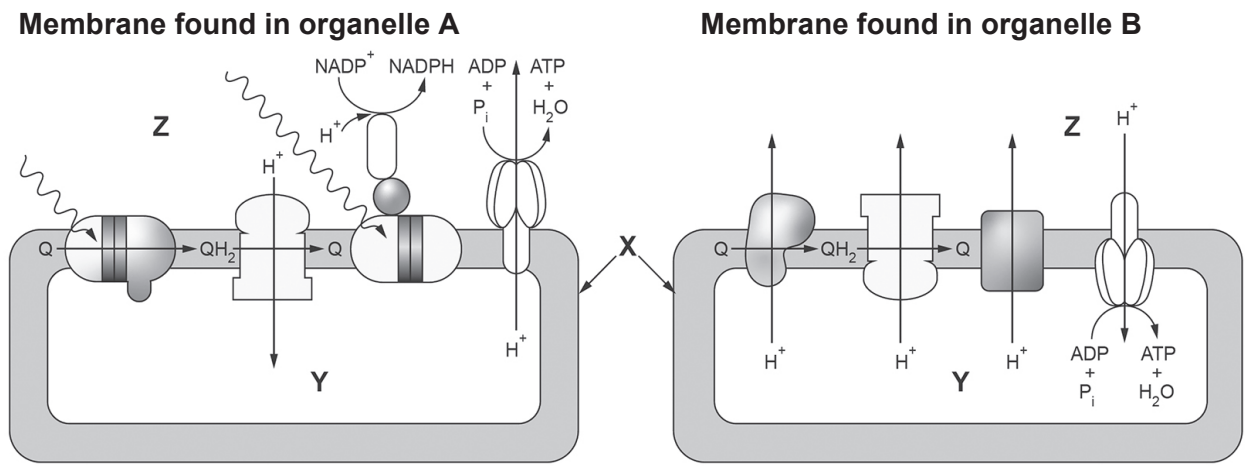


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- (b) **Image 3.2** shows diagrammatic representations of membranes found in two organelles where ATP synthesis takes place.

Image 3.2



- (i) Name the organelles in which the membranes (shown in **Image 3.2**) would be found. [1]

A **B**

- (ii) In **Image 3.2**, **X**, **Y** and **Z** represent membranes and compartments found in organelles **A** and **B**.

Complete the Table 3.3 to name the membranes and compartments represented in **Image 3.2** for each organelle. [3]

Table 3.3

Name of membrane / compartment			
Letter	Part represented in image	Organelle A	Organelle B
X	membrane		
Y	compartment enclosed by membrane		
Z	compartment surrounding membrane		



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(iii) Describe how the components in the membrane of organelle **B** are involved in the synthesis of ATP by chemiosmosis. [5]

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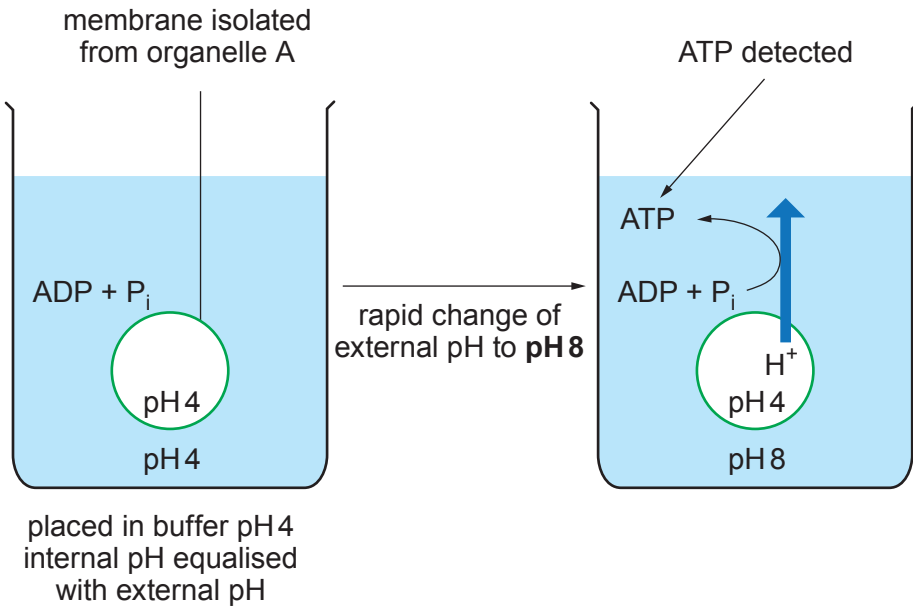
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The experiment shown in **Image 3.4** is considered to be evidence supporting the chemiosmotic hypothesis. It was carried out on membranes isolated from organelles of type **A** in **Image 3.2** and made into vesicles. The isolated membranes were placed in buffer solutions, the lower the pH the higher the concentration of protons.

Image 3.4



(c) Explain why the experiment shown in **Image 3.4** supports the chemiosmotic theory. [4]

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